EcoCode: Green-Software Analysis Platform

Project Proposal for UCS503P
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1 Project Idea

EcoCode is a green-software analysis platform for Java projects. It is intended to analyse complete multi-file repositories, identify source-code patterns that may lead to unnecessary computation or resource usage, and provide evidence-based recommendations. The analysis is deterministic and rule-based rather than AI-dependent.

2 Problem Statement

Software can waste CPU time, memory, I/O, database operations and developer effort because inefficient patterns are difficult to identify consistently in large code bases. Existing manual review is time-consuming and may miss cross-file relationships. EcoCode aims to automate the first stage of this review by building a project model, extracting method relationships and detecting potential resource-wasting patterns.

3 Proposed Solution

The system will incrementally provide:

1. whole-project Java indexing and AST parsing;
2. class, method and method-call modelling;
3. lightweight symbol resolution and call-graph construction;
4. deterministic static-analysis rules (EC101–EC108);
5. later-stage build execution, runtime profiling, evidence validation, recommendations and version comparison.

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4 Development Methodology

The project follows an Agile iterative and incremental model. Early iterations deliver a usable static-analysis pipeline before runtime intelligence and product integration are added. This allows each stage to be tested on real open-source repositories and refined before the next stage.

5 Initial Technology Stack

- Java 21 and Gradle
- JavaParser for AST analysis
- JUnit 5 for automated tests
- Planned later-stage tools: Docker, JFR/JMC, JGit and a web dashboard

6 Expected Outcome

A working prototype that can accept a Java project path, index and parse the repository, construct a project model and call graph, execute deterministic rules, and report source-level static findings. Rule accuracy will be validated and refined using multiple repositories before being treated as final recommendations.